Annotation:

This project investigates the application of machine learning for predicting heart disease using the UCI Cleveland Heart Disease dataset (303 observations). We implemented and compared four models: Logistic Regression, K-Nearest Neighbors (KNN), Decision Trees, and Support Vector Machines (SVM).
The **SVM model** achieved the highest performance with a test accuracy of **0.87** and a recall of **0.91**, which is critical for medical diagnostics. Key predictors identified include oldpeak (ST depression) and thalach (maximum heart rate). Our findings demonstrate that while simple models like Logistic Regression are robust (CV accuracy of 0.86), feature scaling and kernel-based methods provide the necessary precision for clinical decision-making.

Introduction:

Predicting heart disease is a vital task in healthcare to reduce mortality through early intervention. This study utilizes the Cleveland dataset, consisting of 14 clinical features and 303 patient records. The problem is framed as a **binary classification** task: identifying the presence (1) or absence (0) of heart disease.

Hypothesis: We hypothesize that physiological markers of cardiac stress, specifically oldpeak and thalach, will be more statistically significant than demographic factors like age or sex.

EDA summary:

Our exploratory analysis revealed several critical patterns:

- **Target Balance:** The dataset is relatively balanced (54% healthy, 46% diseased), reducing the need for synthetic oversampling.
- **Correlation Analysis:** The heatmap showed strong correlations between the target and cp (chest pain type), exang(exercise-induced angina), and oldpeak.
- **Feature Distributions:** Healthy patients generally exhibited higher thalach (max heart rate) values, while diseased patients showed significantly higher oldpeak values. A key finding was that cholesterol levels

(chol) showed high variance and overlap between groups, making it a less reliable individual predictor.

Methodology:

We selected four models to cover different mathematical approaches:

1. **Logistic Regression:** Chosen as the baseline for its interpretability and efficiency in binary tasks.
2. **KNN:** Selected to capture local patterns in the feature space.
3. **Decision Tree:** Used to identify non-linear decision boundaries and feature importance.
4. **SVM (RBF Kernel):** Selected for its ability to handle high-dimensional data and non-linear relationships.

Preprocessing Pipeline:

- **Missing Values:** Imputed ca with median and thal with mode.
- **Encoding:** One-hot encoding applied to categorical variables (cp, restecg, etc.).
- **Scaling:** StandardScaler was applied to all numerical features. This was mandatory, as our initial tests showed KNN and SVM fail significantly on unscaled data.

Model	Test Accuracy	Precision (Class 1)	Recall (Class 1)	F1-Score	Mean CV Accuracy
Logistic Regression	0.84	0.82	0.88	0.85	**0.8595**
KNN	0.82	0.82	0.84	0.83	0.8265
SVM	**0.87**	**0.85**	**0.91**	**0.88**	0.8428
Decision Tree	0.75	0.75	0.78	0.77	0.7437

The **SVM** is the best model for this use case because its high **Recall (0.91)** minimizes "False Negatives"—instances where a sick patient is incorrectly labeled as healthy.

Error Analysis:

A detailed review of misclassifications revealed that:

1. **Scaling Sensitivity:** Without StandardScaler, KNN accuracy dropped below 65% because the chol feature (range 126–564) dominated the oldpeak feature (range 0–6).
2. **Edge Cases:** Most errors occurred in patients with "asymptomatic" chest pain (cp type 4) who otherwise had normal vitals. The models struggled when clinical markers contradicted the typical disease profile.
3. **Overfitting:** The Decision Tree showed high training accuracy but dropped significantly on the test set, indicating it was "memorizing" the small dataset rather than learning general patterns.

Reflection:

The project confirmed our hypothesis regarding oldpeak, but showed that heart disease is a complex interplay of factors rather than a single-marker diagnosis.

Limitations: The sample size (303) is extremely small for modern machine learning. Our test results showed high variance; changing the random_state could swing accuracy by ±5%.

Improvement: In a future iteration, we would use **Nested Cross-Validation** to get a more "honest" estimate of model performance and explore **Random Forests** to reduce the variance seen in our single Decision Tree.